

(Unofficial) RPI Beamer Presentation Template

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Presentation Overview

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Introduction

A subtitle

What is this?

A template for beamer presentations. I made this template following RPI's official powerpoint template and logos [1].

Why did I do this?

I couldn't find an existing one and I didn't want to copy over citations manually.

Before We Start

- This project uses the fontspec package, which is not supported by the default compiler backend on Overleaf.
- You can change this by following the two steps:

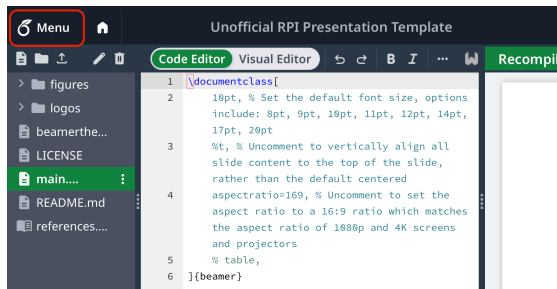


Figure: Open the menu tab

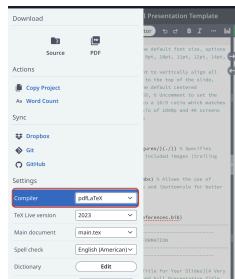


Figure: Set compiler to XeLaTeX

Style Notes

- Use of columns is encouraged.¹
 - Why? because I designed the template with it in mind. minipages works well in most cases, but columns more beamer-native.
- I tried to use relative values for margins and header/sizes where I could. It should scale well with different page sizes.
- Just use `ref` for references. it's pretty annoying to get `cleveref` to work with beamer [2], and it doesn't look like there will be a nice way any time soon.

¹I used to say minipages, but I changed my mind.

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Table of Contents

- You already saw the table of contents.
- By default, it is in one column. If you want to divide it to multiple columns, you can use `minipage` or `columns` [3]
- To make use of this, you need to use sections, which should be placed outside of the frame [4].
- Use `AtBeginSection` to automatically show the table of contents at the beginning of each section.

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Using lists

Simple example of using lists.

- Item 1.

- Item 2.

1. Enumerate 1

2. Enumerate 2

I think this looks pretty similar to how the powerpoint one looks.

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Using figures

Here is an example of a figure.

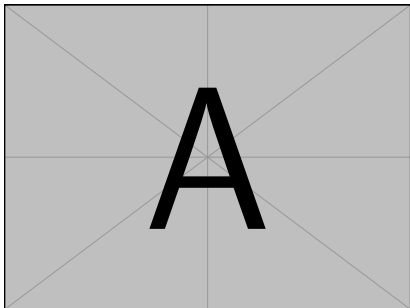


Figure: A subfigure

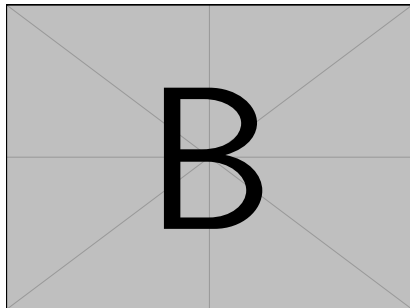


Figure: Another subfigure

Figure: Example of a figure side by side.

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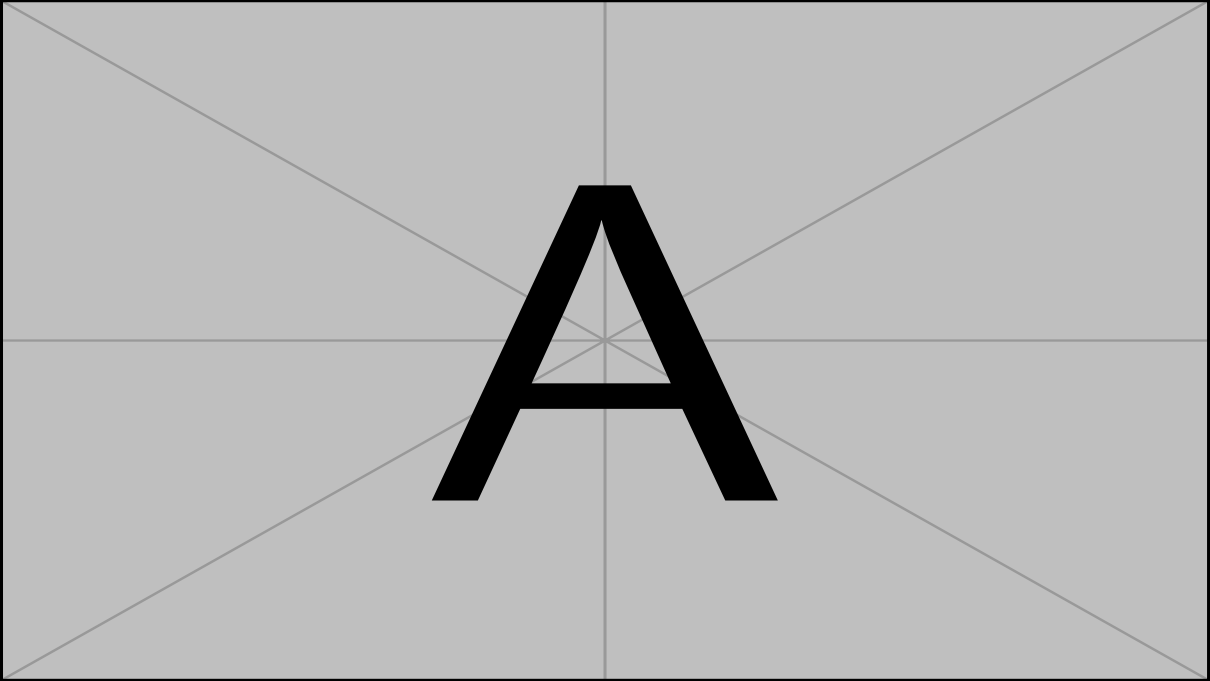
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Using a full screen image

Next slide shows how to use an image in full screen. I got this from [here](#)



A large, bold, black capital letter 'A' is centered on a light gray background. The background features a faint grid of thin gray lines, including a vertical line, a horizontal line, and two diagonal lines forming an 'X' shape. The letter 'A' is composed of thick black strokes, with a horizontal crossbar and two slanted legs. The overall composition is clean and minimalist.

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A Table

Col 1	Col 2	Col 3
a	b	c
d	e	f

Table: An example table

FINDINGS

- This is indeed a table.
- Use minipage to put two things side by side, just like the images.
- You can also use columns, but it's not necessary [5].

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Textboxes

First Block

- First item of first block.
- Second item of first block.
- Notice that the next block matches the height of this one.
- I am also changing the indentation of bullets so it looks better in the boxes.

Second Block

- First item of second block.
 - first sub item of second block.

Third Block

This one doesn't have a list and doesn't match the height!

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Equations

$$\mathcal{L}_{\text{CE}} = \sum_{c=1}^C \log \frac{\exp(x_{n,c})}{\sum_{i=1}^C \exp(x_{n,i})} \quad (1)$$

- Equation 1 shows the cross-entropy loss.
- You can also use inline equations like this: $x_{n,c}$.

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Definitions

Definition

This is a definition.

- It can also be used to define something else.
- It's just a specialblock with the title "Definition".
- If you don't like the colors, you can change block title and block body of setbeamercolor in the .sty file.

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Using BibLaTeX

- Add a bibliography with `\addbibresource` and render it with `\printbibliography`.
- Use `\citep` for parenthetical citations, such as these resources [6]–[8].
- Use `\citet` when the authors are part of the sentence, as in Anglim [6].
- Add `[allowframebreaks]` to the references frame when the bibliography needs multiple slides.
- Use `\bfcite` (mnemonic: blank footcite) to print a full citation at the bottom of the frame without a footnote superscript.
 - Inside a column or minipage, cite with `\citep` or `\citet`.
 - Place the matching `\bfcite` after the columns or minipage so the citation spans the full frame.

[7] C. Fiandrino, “Answer to “Design a custom Beamer theme from scratch”,” (Nov. 2013), [Online]. Available: <https://tex.stackexchange.com/a/146682>

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Footer Title Option/Override

- The default behavior is to use the shorttitle (the stuff defined in [] after \title)
- You can change this by specifying an option when importing the template (like \usetheme[<OPTION_HERE>]{rpi}). Valid options are:
 - footertitlemode=onlysection
 - footertitlemode=sectionandsubsection
 - footertitlemode=onlysubsection
- OR you can override it by using \overridefootertitle{<YOUR_TITLE_HERE>}. This will override the title for the rest of the presentation, until you call \clearoverridefootertitle.

Another slide with custom footer title

Placeholder slide to show resetting the footer title

References I

- [1] RPI, “Home | Strategic Communications and External Relations,” (), [Online]. Available: <https://scer.rpi.edu/>.
- [2] A.-M. Aldaoudeyeh, “Hyperlinks do not work with cleveref + beamer,” (Mar. 2019), [Online]. Available: <https://tex.stackexchange.com/q/481022>.
- [3] samcarter_is_at_topanswers.xyz, “Answer to “Beamer: Vertical alignment of multi-column ToC”,” (Oct. 2013), [Online]. Available: <https://tex.stackexchange.com/a/136082>.
- [4] camp_a, “Answer to “Atbeginsection returns missing number treated as zero”,” (Sep. 2022), [Online]. Available: <https://tex.stackexchange.com/a/657724>.
- [5] M. Scharrer, “Answer to “Minipage, columns, tabular: Differences in applicability?”” (Nov. 2011), [Online]. Available: <https://tex.stackexchange.com/a/34037>.
- [6] J. Anglim, “Answer to “Image on full slide in beamer package”,” (Oct. 2010), [Online]. Available: <https://tex.stackexchange.com/a/3927>.

References II

- [7] C. Fiandrino, “Answer to “Design a custom Beamer theme from scratch”,” (Nov. 2013), [Online]. Available: <https://tex.stackexchange.com/a/146682>.
- [8] J. Kormylo, “Answer to “Make tikz nodes flush with slide edges in beamer title slide”,” (Oct. 2017), [Online]. Available: <https://tex.stackexchange.com/a/395544>.